

LA Lab - 9

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Aim : Construct a two-LAN network in Packet Tracer and statically assign IP addresses, subnet masks, and default gateways on every device.

Theory :

1. Introduction to Cisco Packet Tracer

Cisco Packet Tracer is a widely used network simulation tool that helps in designing, configuring, and troubleshooting network topologies. It is more than just a diagramming tool, as it replicates the behavior of real networking devices using Cisco IOS (Internetwork Operating System).

This software allows users to perform command-line interface (CLI) configurations in a safe virtual environment without affecting real hardware. It also provides a visual way to observe how data packets travel across network devices and connections.

2. Networking Terminology & Device Details

Key Hardware Components

- Router (Layer 3 Device):
A router functions like a delivery system that directs data between different

networks using IP addresses. In this setup, the 2811 router model is used, which supports additional modules like WIC cards for serial communication.

- Switch (Layer 2 Device):

A switch manages communication within a local network by forwarding data based on MAC addresses. The 2960 switch is a managed device, allowing configuration of features such as VLANs and security settings.

- Server:

A server is a dedicated system that provides services such as DHCP, DNS, or web hosting to other devices connected in the network.

Core Networking Concepts

- Subnetting:

Subnetting is the technique of dividing a larger network into smaller sub-networks. This improves network efficiency, reduces unnecessary traffic, and enhances security.

- Default Gateway:

The default gateway is the IP address of the router interface that allows devices in a network to communicate with other networks. Without it, communication is limited only within the local network.

- DHCP (Dynamic Host Configuration Protocol):

DHCP automatically assigns IP addresses to devices, eliminating the need for manual configuration and reducing errors.

- VLAN (Virtual Local Area Network):

VLANs allow logical grouping of devices. Even if devices are physically connected to the same switch, they behave as separate networks unless routing is enabled between them.

3. Implementation Details

Subnetting Logic (Task 2)

The network 192.168.100.0/24 was divided using a /26 subnet mask.

- Borrowed bits: 2 (from /24 to /26)
- Number of subnets: 4
- Usable hosts per subnet: 62

This division helps in better utilization of IP addresses and minimizes broadcast traffic.

Static Routing (Task 3)

A WAN connection was created using serial cables. Since routers were not directly connected to all networks, static routing was configured.

Concept:

If a router needs to send data to a network it does not directly know, it forwards the data to the next router (next hop), which then continues the process.

DHCP Configuration (Task 4)

Router R1 was configured to dynamically assign IP addresses.

- Certain IP addresses (.1 to .9) were reserved for important devices such as routers.
- Lease time was defined to control how long a device can use an assigned IP address.

This automation simplifies network management and avoids IP conflicts.

Layer 2 Security (Task 5)

Port security was implemented using the sticky MAC feature.

- Sticky Learning: The switch automatically learns and stores the MAC address of the first connected device.

- Violation Action: If an unknown device connects, the port is disabled to prevent unauthorized access.

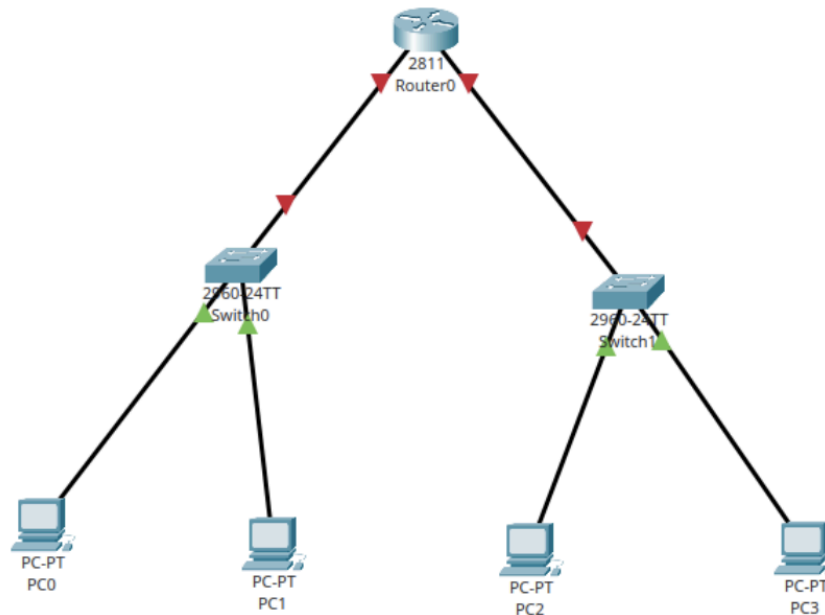
This ensures only trusted devices can connect to the network.

4. Real-World Applications

- ISP and Branch Connectivity:
Static routing and serial connections simulate how branch offices connect to a central network.
 - Network Protection:
Port security prevents unauthorized devices from being connected, protecting the network from potential disruptions or attacks
 - Enterprise Security:
VLANs are used to separate different departments, such as guest networks and internal company networks, improving security.
-

Output :

Task 1 :



```
255K bytes of non-volatile configuration memory.
249856K bytes of ATA System CompactFlash 0 (Read/Write)

--- System Configuration Dialog ---

Would you like to enter the initial configuration dialog? [yes/no]: no

Press RETURN to get started!

Router>enable
Router#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#interface FastEthernet0/0
Router(config-if)#ip address 192.168.10.1 255.255.255.0
Router(config-if)#no shutdown

Router(config-if)#
%LINK-5-CHANGED: Interface FastEthernet0/0, changed state to up

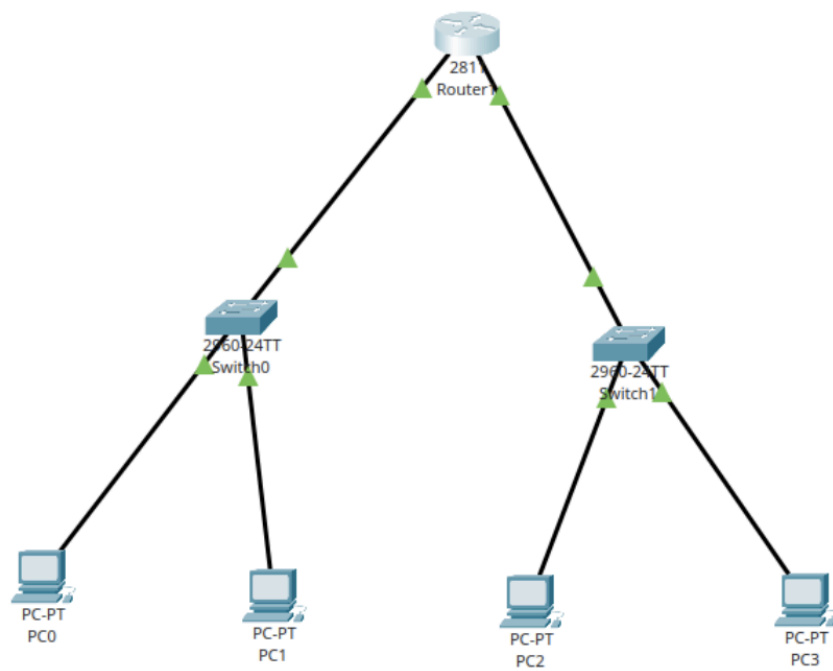
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/0, changed state to up
exit
Router(config)#interface FastEthernet0/1
Router(config-if)#ip address 192.168.20.1 255.255.255.0
Router(config-if)#no shutdown

Router(config-if)#
%LINK-5-CHANGED: Interface FastEthernet0/1, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/1, changed state to up
exit
Router(config)#
```

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PC0

Physical Config **Desktop** Programming Attributes

IP Configuration X

Interface FastEthernet0

IP Configuration

☐ DHCP ☒ Static

IPv4 Address 192.168.10.2

Subnet Mask 255.255.255.0

Default Gateway 192.168.10.1

DNS Server 0.0.0.0

IPv6 Configuration

PC1

Physical Config **Desktop** Programming Attributes

IP Configuration X

Interface FastEthernet0

IP Configuration

☐ DHCP ☒ Static

IPv4 Address 192.168.10.3

Subnet Mask 255.255.255.0

Default Gateway 192.168.10.1

DNS Server 0.0.0.0

IPv6 Configuration

PC2

Physical Config **Desktop** Programming Attributes

IP Configuration X

Interface FastEthernet0

IP Configuration

☐ DHCP ☒ Static

IPv4 Address 192.168.20.2

Subnet Mask 255.255.255.0

Default Gateway 192.168.20.1

DNS Server 0.0.0.0

IPv6 Configuration

PC3

Physical Config **Desktop** Programming Attributes

IP Configuration X

Interface FastEthernet0

IP Configuration

☐ DHCP ☒ Static

IPv4 Address 192.168.20.3

Subnet Mask 255.255.255.0

Default Gateway 192.168.20.1

DNS Server 0.0.0.0

IPv6 Configuration

Task 2 :

```
exit
Router(config)#interface FastEthernet0/1
Router(config-if)#ip address 192.168.20.1 255.255.255.0
Router(config-if)#no shutdown

Router(config-if)#
%LINK-5-CHANGED: Interface FastEthernet0/1, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/1, changed state to up
exit
Router(config)#interface Fa0/0
Router(config-if)#ip address 192.168.100.1 255.255.255.192
Router(config-if)#exit
Router(config)#interface Fa0/1
Router(config-if)#ip address 192.168.100.65 255.255.255.192
Router(config-if)#exit
Router(config)#
```

☐ Top

PC0

Physical Config Desktop Programming Attributes

IP Configuration X

Interface FastEthernet0

IP Configuration

☐ DHCP ☒ Static

IPv4 Address 192.168.10.2

Subnet Mask 255.255.255.192

Default Gateway 192.168.100.1

DNS Server 0.0.0.0

PC1

Physical Config Desktop Programming Attributes

IP Configuration X

Interface FastEthernet0

IP Configuration

☐ DHCP ☒ Static

IPv4 Address 192.168.10.3

Subnet Mask 255.255.255.192

Default Gateway 192.168.100.1

DNS Server 0.0.0.0

PC2

Physical Config **Desktop** Programming Attributes

IP Configuration X

Interface FastEthernet0

IP Configuration

☐ DHCP ☒ Static

IPv4 Address 192.168.20.2

Subnet Mask 255.255.255.192

Default Gateway 192.168.100.65

DNS Server 0.0.0.0

PC3

Physical Config **Desktop** Programming Attributes

IP Configuration X

Interface FastEthernet0

IP Configuration

☐ DHCP ☒ Static

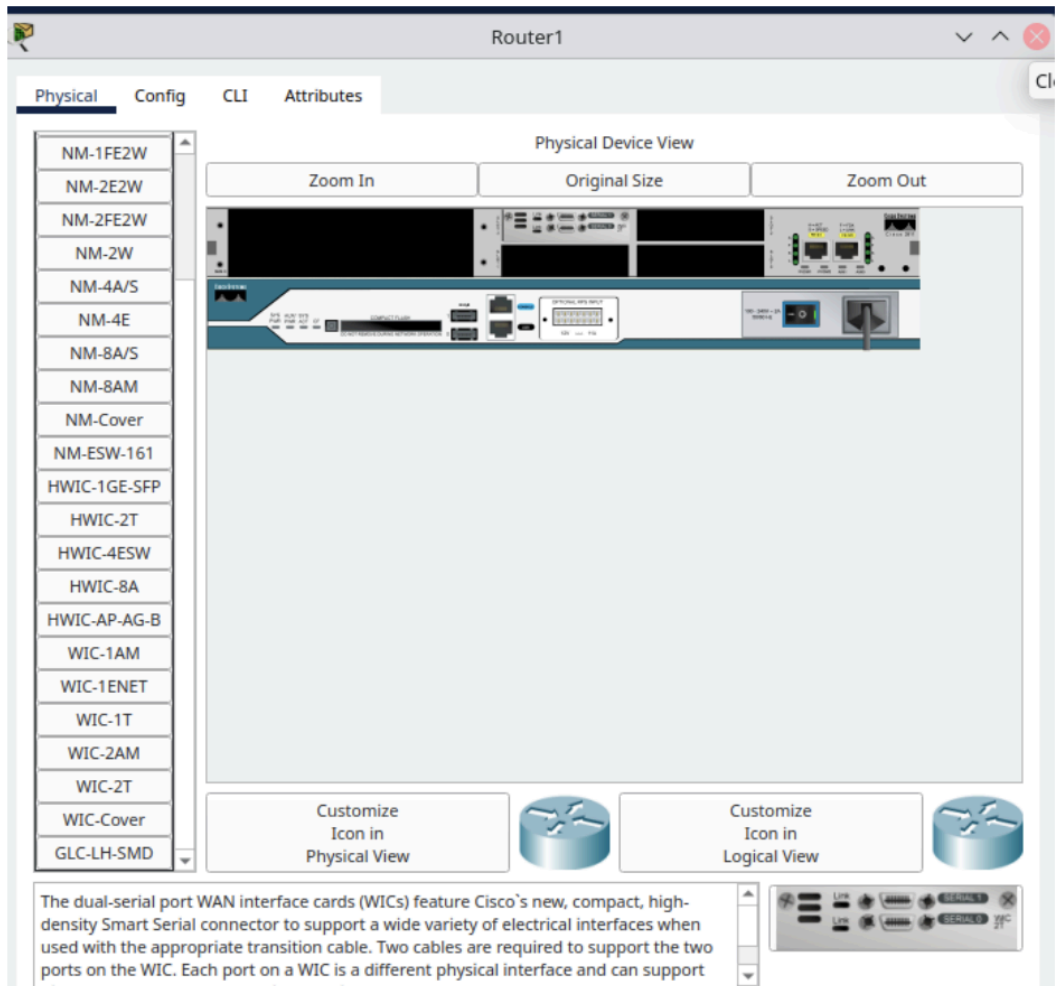
IPv4 Address 192.168.20.3

Subnet Mask 255.255.255.192

Default Gateway 192.168.100.65

DNS Server 0.0.0.0

Task 3 :



IOS Command Line Interface

```
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/0, changed state to up
exit
Router(config)#interface FastEthernet0/1
Router(config-if)#ip address 192.168.100.65 255.255.255.192
Router(config-if)#no shutdown

Router(config-if)#
%LINK-5-CHANGED: Interface FastEthernet0/1, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/1, changed state to up
exit
Router(config)#ip route 192.168.30.0 255.255.255.0 10.0.0.2
Router(config)#ip route 192.168.30.0 255.255.255.0 10.0.0.2
Router(config)#exit
Router#
%SYS-5-CONFIG_I: Configured from console by console
show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is not set

10.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C    10.0.0.0/30 is directly connected, Serial0/3/0
L    10.0.0.1/32 is directly connected, Serial0/3/0
S    192.168.30.0/24 [1/0] via 10.0.0.2
     192.168.100.0/24 is variably subnetted, 4 subnets, 2 masks
C    192.168.100.0/26 is directly connected, FastEthernet0/0
L    192.168.100.1/32 is directly connected, FastEthernet0/0
C    192.168.100.64/26 is directly connected, FastEthernet0/1
L    192.168.100.65/32 is directly connected, FastEthernet0/1

Router#
```

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Router2



Physical Config **CLI** Attributes

IOS Command Line Interface

```
Router(config-if)#ip address 192.168.30.1 255.255.255.0
Router(config-if)#no shutdown

Router(config-if)#
%LINK-5-CHANGED: Interface FastEthernet0/0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/0, changed state to up
exit
Router(config)#ip route 192.168.20.0 255.255.255.0 10.0.0.1
Router(config)#ip route 192.168.20.0 255.255.255.0 10.0.0.1
Router(config)#ip route 192.168.10.0 255.255.255.0 10.0.0.1
Router(config)#show ip route
      ^
% Invalid input detected at '^' marker.

Router(config)#exit
Router#
%SYS-5-CONFIG_I: Configured from console by console
show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is not set

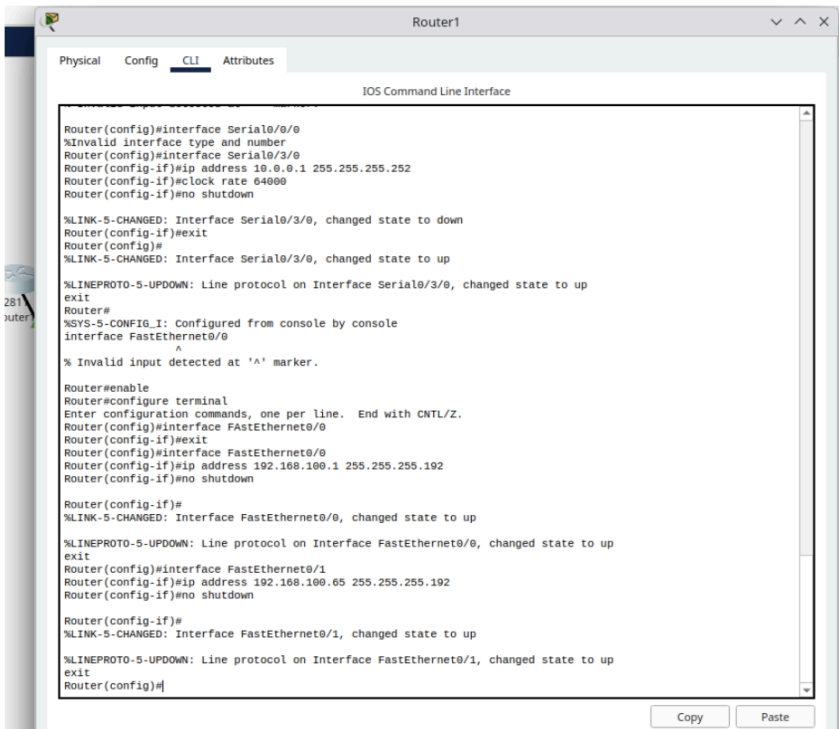
10.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C    10.0.0.0/30 is directly connected, Serial0/3/0
L    10.0.0.2/32 is directly connected, Serial0/3/0
S    192.168.10.0/24 [1/0] via 10.0.0.1
S    192.168.20.0/24 [1/0] via 10.0.0.1
     192.168.30.0/24 is variably subnetted, 2 subnets, 2 masks
C    192.168.30.0/24 is directly connected, FastEthernet0/0
L    192.168.30.1/32 is directly connected, FastEthernet0/0

Router#
```

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Task 4 :



Router1

Physical Config CLI Attributes

IOS Command Line Interface

```
Router(config)#interface Serial0/0/0
%Invalid interface type and number
Router(config)#interface Serial0/3/0
Router(config-if)#ip address 10.0.0.1 255.255.255.252
Router(config-if)#clock rate 64000
Router(config-if)#no shutdown

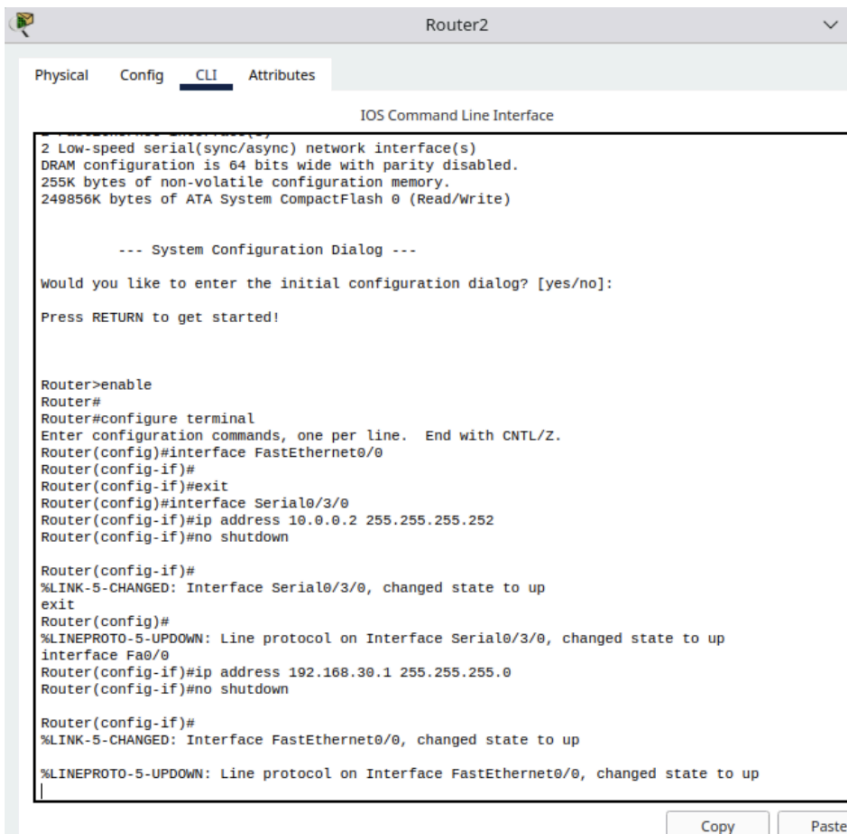
%LINK-5-CHANGED: Interface Serial0/3/0, changed state to down
Router(config-if)#exit
Router(config)#
%LINK-5-CHANGED: Interface Serial0/3/0, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface Serial0/3/0, changed state to up
exit
Router#
%SYS-5-CONFIG_I: configured from console by console
interface FastEthernet0/0
^
% Invalid input detected at '^' marker.

Router#enable
Router#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#interface FastEthernet0/0
Router(config-if)#exit
Router(config)#interface FastEthernet0/0
Router(config-if)#ip address 192.168.100.1 255.255.255.192
Router(config-if)#no shutdown

Router(config-if)#
%LINK-5-CHANGED: Interface FastEthernet0/0, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/0, changed state to up
exit
Router(config)#interface FastEthernet0/1
Router(config-if)#ip address 192.168.100.65 255.255.255.192
Router(config-if)#no shutdown

Router(config-if)#
%LINK-5-CHANGED: Interface FastEthernet0/1, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/1, changed state to up
exit
Router(config)#
```

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Router2

Physical Config CLI Attributes

IOS Command Line Interface

```
2 Low-speed serial(sync/async) network interface(s)
DRAM configuration is 64 bits wide with parity disabled.
255K bytes of non-volatile configuration memory.
249856K bytes of ATA System CompactFlash 0 (Read/Write)

--- System Configuration Dialog ---

Would you like to enter the initial configuration dialog? [yes/no]:

Press RETURN to get started!

Router>enable
Router#
Router#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#interface FastEthernet0/0
Router(config-if)#
Router(config-if)#exit
Router(config)#interface Serial0/3/0
Router(config-if)#ip address 10.0.0.2 255.255.255.252
Router(config-if)#no shutdown

Router(config-if)#
%LINK-5-CHANGED: Interface Serial0/3/0, changed state to up
exit
Router(config)#
%LINEPROTO-5-UPDOWN: Line protocol on Interface Serial0/3/0, changed state to up
interface Fa0/0
Router(config-if)#ip address 192.168.30.1 255.255.255.0
Router(config-if)#no shutdown

Router(config-if)#
%LINK-5-CHANGED: Interface FastEthernet0/0, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/0, changed state to up
|
```

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Router1

Physical Config CLI Attributes

IOS Command Line Interface

E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
* - candidate default, U - per-user static route, o - ODR
P - periodic downloaded static route

Gateway of last resort is not set

10.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C 10.0.0.0/30 is directly connected, Serial0/3/0
L 10.0.0.1/32 is directly connected, Serial0/3/0
S 192.168.30.0/24 [1/0] via 10.0.0.2
192.168.100.0/24 is variably subnetted, 4 subnets, 2 masks
C 192.168.100.0/26 is directly connected, FastEthernet0/0
L 192.168.100.1/32 is directly connected, FastEthernet0/0
C 192.168.100.64/26 is directly connected, FastEthernet0/1
L 192.168.100.65/32 is directly connected, FastEthernet0/1

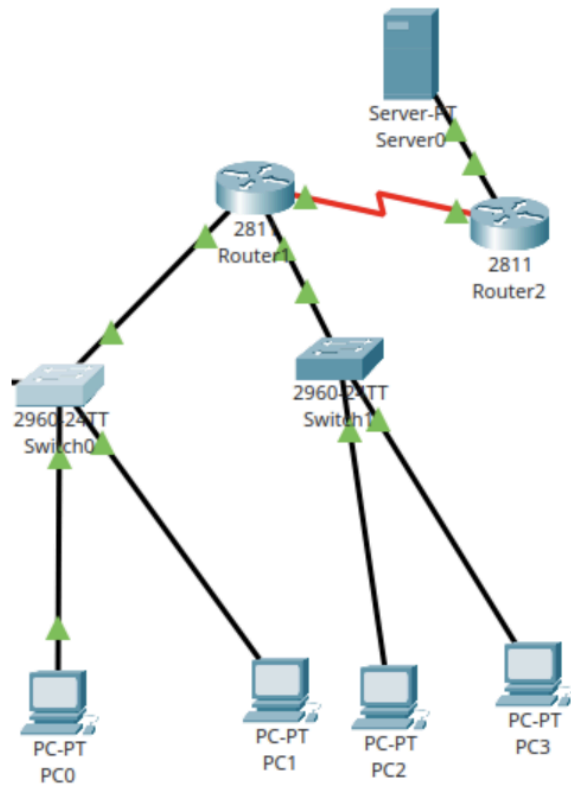
Router#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#ip dhcp excluded-address 192.168.10.1 192.168.10.9
Router(config)#ip dhcp excluded-address 192.168.20.1 192.168.20.9
Router(config)#ip dhcp pool LAN1
Router(dhcp-config)#network 192.168.10.0 255.255.255.0
Router(dhcp-config)#default-router 192.168.10.1
Router(dhcp-config)#dns-server 8.8.8.8
Router(dhcp-config)#lease 7
^
% Invalid input detected at '^' marker.

Router(dhcp-config)#lease 7 0 0
^
% Invalid input detected at '^' marker.

Router(dhcp-config)#exit
Router(config)#ip dhcp pool LAN2
Router(dhcp-config)#network 192.168.20.0 255.255.255.0
Router(dhcp-config)#default-router 192.168.20.1
Router(dhcp-config)#dns-server 8.8.8.8
Router(dhcp-config)#

Router#traceroute 10.0.0.1
Type escape sequence to abort.
Tracing the route to 10.0.0.1

1 10.0.0.2 0 msec 5 msec 9 msec
2 10.0.0.1 1 msec 1 msec 1 msec



Task 5 :



The screenshot shows a network switch interface with tabs for Physical, Config, CLI, and Attributes. The CLI tab is active, displaying the IOS Command Line Interface. The terminal output shows the following commands and their results:

```
Switch>enable
Switch#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
Switch(config)#vlan 10
Switch(config-vlan)#name HR
Switch(config-vlan)#exit
Switch(config)#vlan 20
Switch(config-vlan)#name IT
Switch(config-vlan)#exit
Switch(config)#interface fa0/1
Switch(config-if)#switchport mode access
Switch(config-if)#switchport access vlan 10
Switch(config-if)#exit
Switch(config)#interface fa0/2
Switch(config-if)#switchport mode access
Switch(config-if)#switchport access vlan 20
Switch(config-if)#exit
Switch(config)#interface fa0/24
Switch(config-if)#switchport mode trunk

Switch(config-if)#
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/24, changed state to down

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/24, changed state to up
exit
Switch(config)#interface fa0/1
Switch(config-if)#switchport port-security
Switch(config-if)#switchport port-security maximum 1
Switch(config-if)#switchport port-security violation shutdown
Switch(config-if)#switchport port-security mac-address sticky
Switch(config-if)#
```

Below the terminal output, the command `Switch#show port-security address` is executed, resulting in the following table:

Secure Mac Address Table				
Vlan	Mac Address	Type	Ports	Remaining Age (mins)
10	000C.CF6E.7267	SecureSticky	Fa0/1	-

Additional information displayed at the bottom of the output:

```
Total Addresses in System (excluding one mac per port) : 0
Max Addresses limit in System (excluding one mac per port) : 1024
Switch#
```

```
IOS Command Line Interface
show port-security interface fa0/1
Port Security          : Enabled
Port Status            : Secure-down
Violation Mode         : Shutdown
Aging Time             : 0 mins
Aging Type             : Absolute
SecureStatic Address Aging : Disabled
Maximum MAC Addresses  : 1
Total MAC Addresses    : 1
Configured MAC Addresses : 0
Sticky MAC Addresses   : 1
Last Source Address:Vlan : 000C.CF6E.7267:10
Security Violation Count : 0

Switch#
%LINK-5-CHANGED: Interface FastEthernet0/1, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/1, changed state to up
show port-security interface fa0/1
Port Security          : Enabled
Port Status            : Secure-up
Violation Mode         : Shutdown
Aging Time             : 0 mins
Aging Type             : Absolute
SecureStatic Address Aging : Disabled
Maximum MAC Addresses  : 1
Total MAC Addresses    : 1
Configured MAC Addresses : 0
Sticky MAC Addresses   : 1
Last Source Address:Vlan : 000C.CF6E.7267:10
Security Violation Count : 0

Switch#
%LINK-5-CHANGED: Interface FastEthernet0/1, changed state to administratively down

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/1, changed state to down
%PM-4-ERR_DISABLE: psecure-violation error detected on Fa0/1, putting Fa0/1 in err-disable
state.
%PORT_SECURITY-2-PSECURE_VIOLATION: Security violation occurred, caused by MAC address
000A.F39A.DB42 on port FastEthernet0/1.
```

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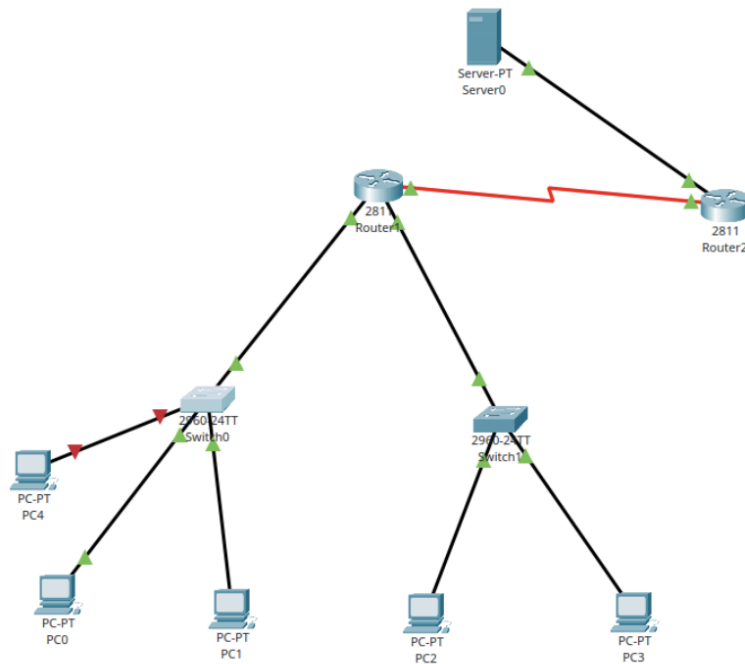
```
Maximum MAC Addresses  : 1
Total MAC Addresses    : 1
Configured MAC Addresses : 0
Sticky MAC Addresses   : 1
Last Source Address:Vlan : 000C.CF6E.7267:10
Security Violation Count : 0

Switch#
%LINK-5-CHANGED: Interface FastEthernet0/1, changed state to administratively down

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/1, changed state to down
%PM-4-ERR_DISABLE: psecure-violation error detected on Fa0/1, putting Fa0/1 in err-disable
state.
%PORT_SECURITY-2-PSECURE_VIOLATION: Security violation occurred, caused by MAC address
000A.F39A.DB42 on port FastEthernet0/1.
show port-security interface fa0/1
Port Security          : Enabled
Port Status            : Secure-shutdown
Violation Mode         : Shutdown
Aging Time             : 0 mins
Aging Type             : Absolute
SecureStatic Address Aging : Disabled
Maximum MAC Addresses  : 1
Total MAC Addresses    : 1
Configured MAC Addresses : 0
Sticky MAC Addresses   : 1
Last Source Address:Vlan : 000A.F39A.DB42:10
Security Violation Count : 1

Switch#
```

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Tables :

Task 2 :

Subnet	Network Address	Usable IP Range	Broadcast Address
A	192.168.100.0	192.168.100.1 – .62	192.168.100.63
B	192.168.100.64	192.168.100.65 – .126	192.168.100.127
C	192.168.100.128	192.168.100.129 – .190	192.168.100.191
D	192.168.100.192	192.168.100.193 – .254	192.168.100.255

Task 3 :

Device	DHCP assigned IP address	Gateway Received
--------	--------------------------	------------------

PC0	192.168.10.2	192.168.100.1
PC1	192.168.10.3	192.168.100.1
PC2	192.168.20.2	192.168.100.65
PC3	192.168.20.3	192.168.100.65

Task 4 :

IP Address	Device
192.168.10.0/24 via 10.0.0.1	PC0
192.168.20.0/24 via 10.0.0.1	PC1
192.168.30.0/24 via 10.0.0.2	PC2

Task 5 :

Observation	Your answer
What happened to the port after violation?	It went into state "err-disabled"
Command to re enable the shutdown port?	no shutdown
Sticky MAC Address Learned (from show port-security) ?	000C.CF6E.7267

Conclusion : In this experiment, we used Cisco Packet Tracer to design and implement a complete network topology in a virtual environment. We applied important networking concepts like subnetting, static routing, DHCP configuration, VLANs, and port security. Overall, this experiment provided hands-on experience in configuring and managing networks, along with a clear understanding of real-world networking practices.
